1. Identity

Document	STAR-PKG-STAR-TDB-EN · v2.1 · 2026-11-01
Package	star-tdb
Version	0.1.1
Capability	star.time.tdb
Maturity	VERIFIED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_tdb

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_tdb-0.1.1-py3-none-any.whl
```

FILE	SHA-256
star_tdb-0.1.1-py3-none-any.whl	2de8a9bca39a46a57c064bca8c6adcb0a3f2e9e3034a681445850297787f2034
star_tdb.py	126fafbabc0d9fd6e288042c8abbb282516c5ac9859485d09a12688d4d642e35
star_tdb-0.1.1-source.zip	1b03346d75f8c1d9d7e37bfe6a484ea49d4c3e8d56cfa20af7a51254d818692f

5. What it does, and its limits

TDB - TT for a TT Julian date, geocentre. Standard library only.

Topocentric terms (observer position), sub-microsecond accuracy, dates outside 1600-2200.

- `tdb_minus_tt(jd_tt)` → `seconds`, USNO Circular 179 (Kaplan 2005) eq. 2.6, 7 terms.
- Accuracy within 10 microseconds of the full Fairhead & Bretagnon series over 1600-2200.
- Outside 1600-2200 raises `ValueError` (no extrapolation beyond stated validity).

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_tdb_0_1_1,
  author  = {Cavazzini, Andrea},
  title   = {{star-tdb}: TDB - TT with the USNO Circular 179 series, verified against ERFA full series within 10 microseconds},
  version = {0.1.1},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_tdb/},
```

```
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts